

Python & Data Analysis Assignment

This assignment is designed to strengthen Python programming skills and introduce real-world data analysis concepts. Complete the Python tasks and perform Exploratory Data Analysis (EDA) on ONE dataset from the provided options.

Part A: Python Programming Tasks

1. Create a list of 20 numbers. Find the maximum, minimum, sum, and average.
2. Print all even and odd numbers from a given list.
3. Create a dictionary containing information and print all keys and values.
4. Write a function that returns the square of a number.
5. Write a program to count vowels in a string.
6. Write a program to find the second largest number in a list.
7. Write a program to remove duplicate elements from a list.

Part B: Exploratory Data Analysis (EDA) Project

Choose ONLY ONE of the following datasets:

- Healthcare Stroke Dataset
- Bengaluru House Prices Dataset
- MBA Supply Chain Dataset

Mandatory EDA Tasks

1. Load the dataset and display the first 5 rows.
2. Display shape, columns, data types, and dataset information.
3. Generate descriptive statistics using describe().
4. Check and analyze missing values.
5. Check duplicate records.
6. Identify numerical and categorical features.
7. Create at least 2 histograms.
8. Create at least 2 countplots/bar plots.
9. Create at least 1 boxplot.
10. Create at least 1 scatter plot.
11. Create a correlation heatmap.
12. Perform one additional visualization of your choice.
13. Write observations below every visualization.
14. Write at least 10 insights/conclusions from your analysis.

Required Technologies

Use Python, Pandas, NumPy, Matplotlib, and Seaborn.

Submission Requirements

1. Create a GitHub repository for this assignment.
2. Upload your Jupyter Notebook (.ipynb) file.
3. Upload the dataset used for analysis.
4. Upload a text file named learning_summary.txt.

learning_summary.txt Requirements

Write a summary describing:

- What you understood from the dataset.
- What challenges you faced during the analysis.
- Which visualizations were most useful.
- What insights you discovered.
- What you learned about EDA and data analysis.
- Minimum length: 300 words.